

CleanSort AI — Intelligent Waste Classification using Machine Learning

Course: Machine Learning — BSSE

Section: C

Group Members:

Names	Roll Numbers	Assigned Role
Ehtasham Maqbool	231400033	Project Lead
Ahmad Khan	231400101	Data Collection & Preprocessing
M. Bilal	231400038	Evaluation Metrics & Testing
M. Saqib	221370198	UI/Deployment & Documentation Lead

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Section 2 — Problem Statement

Waste management is one of the major environmental challenges in urban areas, particularly in developing countries where proper waste segregation is often neglected. Improper disposal of recyclable and non-recyclable waste leads to environmental pollution, inefficient recycling processes, increased landfill usage, and higher waste management costs. Manual waste classification is labor-intensive, time-consuming, and prone to human error, making it unsuitable for large-scale implementation . To address this problem, this project proposes a **Machine Learning-based Intelligent Waste Classification System** that automatically classifies waste images into six categories: **cardboard, glass, metal, paper, plastic, and trash**. The problem is formulated as a **multi-class classification problem** in Machine Learning. The system aims to improve waste segregation efficiency and support sustainable recycling practices using traditional Machine Learning techniques.

Section 3 — Objectives

- Develop a multi-class Machine Learning classifier for waste image classification.
- Implement **K-Nearest Neighbors (KNN)**, **Naïve Bayes**, and **Logistic Regression** algorithms from scratch without using pre-built classifier libraries.
- Compare the performance of all three models using standard evaluation metrics.
- Achieve at least **80–85% classification accuracy** on the testing dataset.
- Develop a simple prototype interface for real-world waste classification.

Section 4 — Dataset Description

Dataset Name

TrashNet Dataset

Source

<https://www.kaggle.com/datasets/feyzazkefe/trashnet>

Dataset Description

The TrashNet dataset contains approximately **2,500+ RGB waste images** belonging to six classes:

- Cardboard
- Glass
- Metal
- Paper
- Plastic
- Trash

Nature of Features

- Image Data (RGB Images)

After preprocessing, images will be converted into numerical feature vectors using feature extraction techniques.

Data Quality Issues

- Class imbalance

- Background variations
- Lighting variations
- Noise
- Similar appearance among some waste categories

The dataset is publicly available and suitable for academic and research purposes.

Section 5 — Proposed Methodology

The proposed system will implement a traditional Machine Learning pipeline without using Deep Learning models.

Step 1: Data Preprocessing

The following preprocessing techniques will be applied:

- Image resizing
- Image normalization
- Noise reduction
- Grayscale conversion (if required)

The dataset will be divided into:

- **70% Training**
- **15% Validation**
- **15% Testing**

Step 2: Feature Extraction

Since Machine Learning algorithms cannot directly process raw images, handcrafted image features will be extracted.

Feature extraction methods include:

- Histogram of Oriented Gradients (HOG)
- Local Binary Pattern (LBP)
- Color Histogram

The extracted features will be converted into numerical feature vectors.

Step 3: Machine Learning Models

Three Machine Learning algorithms will be implemented **from scratch** and trained using the extracted feature vectors.

Model 1 — K-Nearest Neighbors (KNN)

Reason for Selection:

- Simple implementation
- No training phase
- Effective for image feature classification
- Easy to interpret and compare

Model 2 — Naïve Bayes

Reason for Selection:

- Fast probabilistic classifier
- Simple mathematical implementation
- Good baseline model for comparison

Model 3 — Logistic Regression

Reason for Selection:

- Popular supervised classification algorithm
- Uses gradient descent optimization
- Suitable for multi-class classification using One-vs-Rest strategy
- Easy to evaluate against KNN and Naïve Bayes

Step 4: Model Evaluation

The performance of all three models will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

- Confusion Matrix

The best-performing model will be selected for deployment in the prototype application.

Section 6 — Role Distribution

Project Manager

- Overall project planning
- Progress monitoring
- Team coordination
- Final presentation
- Implementation of KNN

Data Engineer

- Dataset collection
- Image preprocessing
- Feature extraction
- Data preparation

Machine Learning Engineer

- Naïve Bayes
- Logistic Regression
- Model training
- Hyperparameter tuning

Evaluation & Documentation Lead

- Model evaluation
- Result comparison
- Prototype interface
- Documentation
- Report writing

Section 7 — Risk Assessment

Potential Risks

- Limited dataset size
- Class imbalance among waste categories
- Similar visual appearance between different waste classes
- Feature extraction may not capture all discriminative information
- Lower classification accuracy compared to Deep Learning methods

Contingency Plan

If the initial performance is unsatisfactory, the following improvements will be considered:

- Experiment with different handcrafted feature extraction methods
- Optimize model hyperparameters
- Apply feature selection techniques
- Balance the dataset using resampling techniques
- Compare additional traditional Machine Learning algorithms if necessary

References

1. TrashNet Dataset. Kaggle. <https://www.kaggle.com/datasets/feyzazkefe/trashnet>
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3. Cortes, C., & Vapnik, V. (1995). *Support Vector Networks*.
4. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*.
5. Ojala, T., Pietikäinen, M., & Mäenpää, T. (2002). *Multiresolution Gray-Scale and Rotation Invariant Texture Classification using Local Binary Patterns*.